Practical No 15

Aim

Demonstrate the use of a Python debugger (e.g., pdb or an IDE with debugging capabilities) on a sample program with intentional errors. Guide students on setting breakpoints, stepping through code, and examining variable values.

Apparatus / Software Required

- Python Interpreter (Python 3.14.2)

Theory

What is Debugging?

Debugging is the process of identifying and fixing errors in a program. It helps in understanding how the program executes step-by-step.

- **Debugging in IDLE**

IDLE (Python's Integrated Development Environment) provides a simple debugger with features like:

- Breakpoints
- Step-by-step execution
- Variable inspection

- **Key Features in IDLE Debugger**

1. Breakpoints
   - Set by right-clicking on a line → *Set Breakpoint*
   - Marks the line where execution will pause
2. Debugger Window
   - Shows current line execution
   - Displays variables and their values

3. Execution Controls

Button	Function
Go	Run program
Step	Execute next line
Over	Skip function call
Out	Exit function
Quit	Stop execution

Program Code

Output

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Conclusion

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